

Deep Research Prompt: Reusing Dialysis RO-Reject Water in France

You are an expert in health systems, nephrology, and sustainability, supporting a hackathon pitch for reducing the environmental footprint of dialysis in France.

Project goal: Build a slide-ready, evidence-based analysis around one focused idea: **capture and reuse the clean Reverse Osmosis (RO) reject water** from hospital hemodialysis for non-clinical uses (e.g. cooling towers, toilets, irrigation) in French hospitals. The story must be realistic, technically feasible, and tailored to the French context.

Important Context & Constraints

- **Focus:** Only France. Include recent drivers like **Plan Eau 2023** and summer droughts as catalysts for action.
- **Regulatory:** Consider ARS guidelines on non-potable water reuse (since new 2024 regulations govern “eaux impropres à la consommation humaine” for domestic uses ¹ ²). Emphasize that reuse is **for non-clinical purposes only** (never recycling back into dialysis).
- **Funding:** Incorporate French subsidies/grants (e.g. **Les Agences de l'Eau** water agencies) and **CEE** (energy savings certificates) in the business model.
- **Organization:** Use the GHT (Groupement Hospitalier de Territoire) framework to explain scaling from large university hospitals (CHU/CHR) to smaller regional hospitals (inequality/equity considerations).
- **Hardware:** Identify actual equipment needed (e.g., holding tanks, UV-C sterilizers or filtration to prevent biofilms, dual piping for non-potable loop, additional pumps).
- **Sources:** Use only high-quality, recent sources (PubMed, French government/health agency reports, peer-reviewed papers, credible technical reports). Provide citations for all key data.

Methodology (Research Approach)

- Use a structured search approach (PubMed, official French statistics, technical literature).
- Search French sources for hospital/dialysis data (e.g. DREES reports, REIN registry).
- Search PubMed for “dialysis water reuse”, “RO reject water reuse”, “hemodialysis sustainability”, etc.
- Search French regulations (Plan Eau 2023, ARS guidelines, code de la santé publique for EICH).
- Cross-check data across multiple sources. If numbers conflict or are missing, state assumptions clearly.
- Distinguish evidence vs assumptions. Label uncertain or pilot-stage ideas as such.

Required Final Output (Slide-by-Slide)

Write the output as if it will be turned into slides. Use headings and bullet points. Include evidence and concise explanations.

Slide 1: The Problem & The Catalyst

Headline: Why hospital dialysis wastes water – and why action is urgent in France.

Bullet points should include:

- **Water use per session:** ~400 liters of water needed per hemodialysis treatment, of which **50–70% can be RO reject water**.
- **Environmental issue:** This discard is potable water drawn for treatment and then mostly dumped, increasing water stress.
- **French context:** France's *Plan Eau 2023* (launched March 2023) makes water resilience a national priority ³. Exceptional droughts (summer 2022, 2023) led to restrictions, pressuring hospitals to save water.
- **Regulatory push:** New 2024 EICH regulations (décret/arrêté 12 July 2024) ease use of non-potable water for domestic purposes, under ARS oversight ⁴ ⁵. This makes institutional reuse legally feasible (with safeguards).

Slide 2: France Market Size & First Adopters

Provide a compact table or bullet summary. Key data (with sources):

- **Dialysis patients:** ~51,700 in 2022.
- **Modality split:** ~51.9% in-center HD, 25.9% UDM, 15.1% autodialysis (in-center), 6% peritoneal, 1.5% home HD. (About 79% in hospital-based modalities.)
- **Hospitals:** ~2,965 total health establishments; among public hospitals: 184 CHR, 924 CH, etc.
- **Large sites:** Many dialysis centers have 30–100+ patients (the median center has ~25 patients, but large CHU/CHR handle 100+).
- **First adopters:** *Large CHU/CHR and major private clinics* should pilot first: they have higher patient volumes (justifying capex) and existing infrastructure (space, tech teams) to implement reuse.
- **Smaller sites:** Small rural clinics or UDMs (autodialysis units) have low volumes (<10 patients) – not cost-effective for full systems, but can later adopt modular solutions (e.g. small buffer tanks for toilet flushing).

Slide 3: The Technology & Hardware

Ranked options for RO-reject reuse: (score 1–5; higher is better)

- **1. Capture & reuse for non-potable uses** (cooling, toilets, irrigation): *Impact★5, Ease★3, Safety★4, Cost★3, Time-to-pilot★4*. Requires a buffer tank and dual plumbing to segregate RO reject. Needs UV-C or equivalent sanitization to prevent microbial growth. Proven tech in water treatment.
- **2. Onsite storage & treatment (microfiltration/UV):** *Impact★4, Ease★2, Safety★4, Cost★2, Time-to-pilot★3*. More complex (filtration units, regular maintenance), but higher purity of reused water.
- **3. Limited reuse (e.g., only for irrigation or outdoor cleaning):** *Impact★3, Ease★5, Safety★5, Cost★4, Time-to-pilot★5*. Simplest (just collect reject into cistern, use on gardens or external cleaning). Very safe due to distance from people. Lower impact (smaller reuse volume).

Hardware components needed (for option 1, the preferred solution):

- **Holding/storage tanks:** e.g., 1–2 m³ buffer tanks to collect RO reject from dialysis units.

- **Dual-piping network:** Separate non-potable loop (toilets, cooling, irrigation) with backflow prevention.
- **Disinfection:** UV-C or chlorine system to control pathogens in stored water. ARS requires preventing cross-contamination ¹ ⁵.
- **Pumps & controls:** To move water into reused circuits on demand.
- **Monitoring:** Simple sensors for level/flow to optimize use.

(Important: Under no circumstance is reused water sent back into dialysis machines. All reuse is external, non-clinical.)

Slide 4: Evidence & ROI

Present data points with sources and confidence levels. For example:

- **Water saved:** A 100-patient hospital HD center uses ~40,000 sessions/year (~16 million L water/year). If 60% is rejected and reused, ~9.6 million L saved/year (~6,400 tons of water).
- **CO₂ reduction:** Assuming ~0.5 kg CO₂e per m³ of treated potable water (production + heating), saving 9,600 m³ yields ~4.8 tCO₂e saved/year. (Plus energy savings in pumping and heating.)
- **Financials:** Case studies (e.g., a dialysis unit in Lyon, 173 patients) show ROI in ~5–6 years. Smaller pilots report 1–3 year payback due to high water/energy costs.
- **Existing pilots:** International examples (e.g. UK, Netherlands) and one French center (Lyon) have implemented reject water reuse with positive results.
- **Confidence:** High for basic volumes (patient counts, water use per session). Moderate for financials (depends on local water/sewer tariffs and capex). Clearly cite data (e.g. for water usage; for example study).

Slide 5: The Roadmap (Pilot → Scale)

Time Horizon	Key Actions	Expected Impact	Main Blockers	Next Steps to Prove
0–6 months:	- Conduct site audits at 3 large CHUs - Secure ARS approval for pilot reuse (submit EICH usage plan) ⁴ - Apply for Agence de l'Eau subsidies and CEE financing programs	- Identify realistic volumes and costs - Begin small-scale collection (proof of concept)	- Regulatory approval timelines - Capex sourcing (navigate grants)	- Finalize hardware design - Demonstrate water quality safety
~3 years:	- Deploy systems across the GHT: equip medium hospitals and polyclinics - Expand to interested private clinics in large cities (Paris, Lyon, etc.) - Integrate reuse into hospital environmental management	- Save 20–30% of dialysis water nationally - Build expertise and lower unit costs	- Coordination across hospitals - Standardizing procurement and maintenance	- Show multi-site success (e.g., 5+ hospitals) - Adjust regulations if needed

Time Horizon	Key Actions	Expected Impact	Main Blockers	Next Steps to Prove
5–7 years:	- Advanced integration: use reject for heat recovery (preheating incoming water)
- Explore effluent recirculation (using treated spent dialysate in sorbent systems)
- Mandatory water-reuse targets in hospital planning (national policy)	- Maximize sustainability (up to 40%+ water reduction)
- Embed practice as standard care	- Technology readiness for advanced reuse
- Need for regulatory updates	- Long-term performance data
- Policy incentives or regulations

(Note: ARS = Agence Régionale de Santé; GHT = Groupement Hospitalier de Territoire.)

Slide 6: Business Model & Equity Lens

- **Who pays:** Hospital or hospital group (CHU/GHT) makes the capital investment, offset by **Agence de l'Eau** grants and **CEE** subsidies. Maintenance costs could be shared with water utility partners.
- **Who benefits:** The hospital saves on water and sewage bills immediately (lower OPEX) and gains resilience against water restrictions. Patients benefit indirectly from more sustainable care.
- **Partners:**
 - *Public:* ARS (regulatory approval), Agences de l'Eau (funding), ADEME (possibly), local municipalities for irrigation.
 - *Private:* Water service companies (could install/maintain systems under contract), dialysis equipment vendors (for integration).
- **Equity (GHT approach):** Use each GHT as an umbrella: large CHUs lead pilot, then share or lease equipment to smaller member hospitals. Rural or poorer hospitals get support via GHT budgets and national grants.
- Ensure urban vs rural: Town hospitals with small UDM units can use **modular micro-tanks** and simple UV units (lower cost, CEE-supported). Pooling via GHT ensures all regions (Paris/regions) can participate, not just rich centers.

Slide 7: Final Pitch Message (20–30s)

Craft a concise spoken summary. For example:

“In France, a single dialysis session uses ~400 liters of water, half of which is clean RO-reject water. With water shortages mounting (Plan Eau 2023, 2022–24 droughts), we propose capturing that unused water in large hospitals and reusing it for toilets, cooling, and irrigation. This avoids millions of liters of fresh water per year per hospital, cutting costs and CO₂ emissions. Implementation is straightforward: buffer tanks, UV sterilizers, and dual plumbing, fully compliant with ARS regulations. We pilot at big CHUs (using Agence de l'Eau subsidies), then scale through each regional GHT to smaller hospitals. The result: greener dialysis without compromising care. Our solution meets today's crises and builds long-term resilience.”

Output style: Use slide-friendly bullet points. Every key claim or number must be supported by a cited source (e.g. scientific paper, official statistic, case study). Avoid vague language. Focus on the French healthcare/regulatory context and concrete evidence.

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